

Indian Institute of Technology Goa

EE 635 Digital Integrated Circuit Design

Instructor: Nandakumar Nambath

Course Project

Submitted by Group 9:

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1. You have to implement an array multiplier that performs the multiplication of two fourbit numbers, $A = A_3A_2A_1A_0$ and $B = B_3B_2B_1B_0$ as part of this project. The product of A and B is represented by the eight-bit number, $P = P_7P_6P_5P_4P_3P_2P_1P_0$. The circuit diagram of the array multiplier is shown in Fig. 1. You have to follow the steps given below to complete the project.

(a) Implement reference two-input AND, OR, and XOR gates as the basic building blocks of the multiplier. Plot the VTC and find out the propagation delays of the gates assuming that all inputs transition simultaneously

Ans: **For AND GATE**

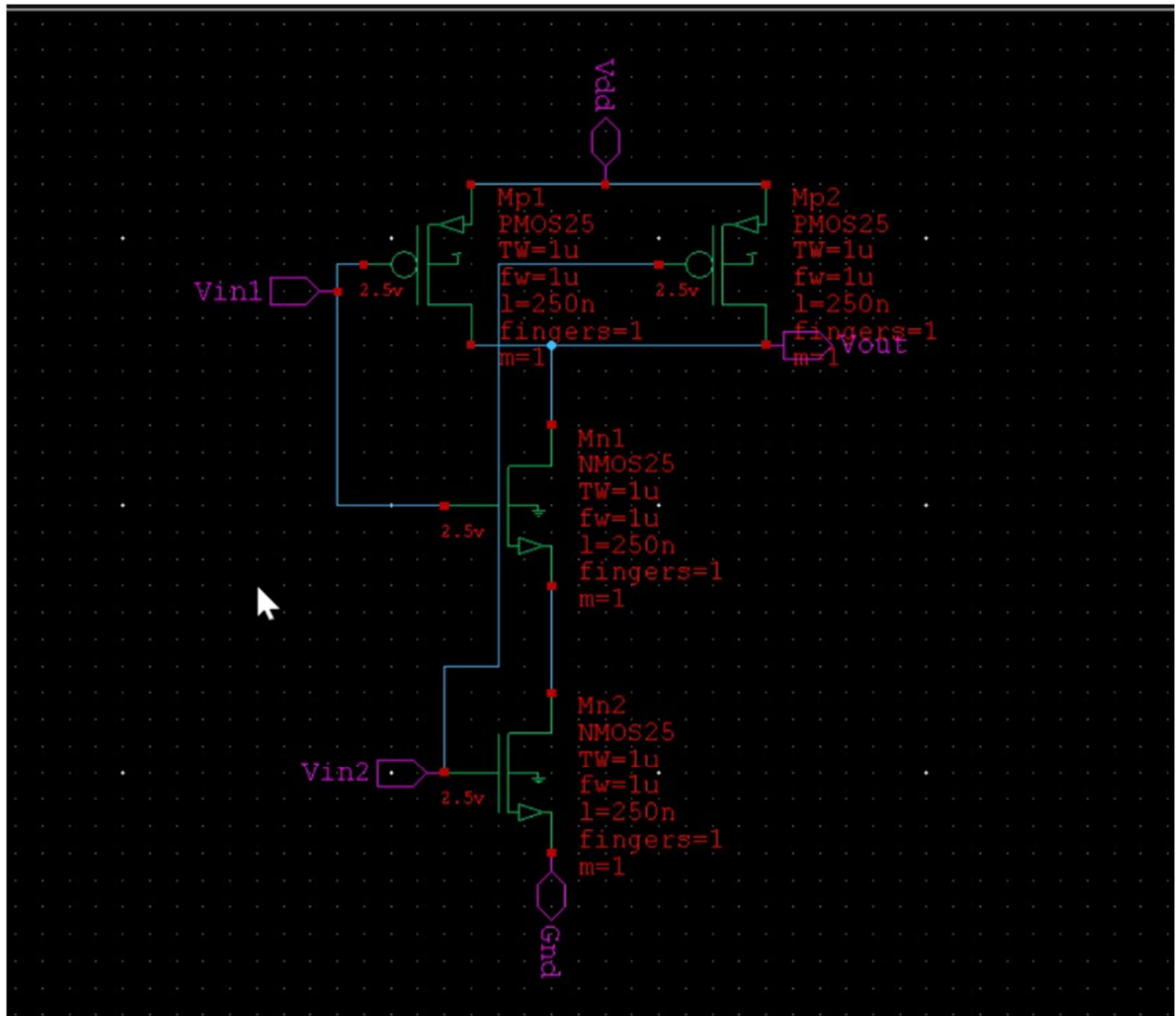


Figure-1 NAND gate

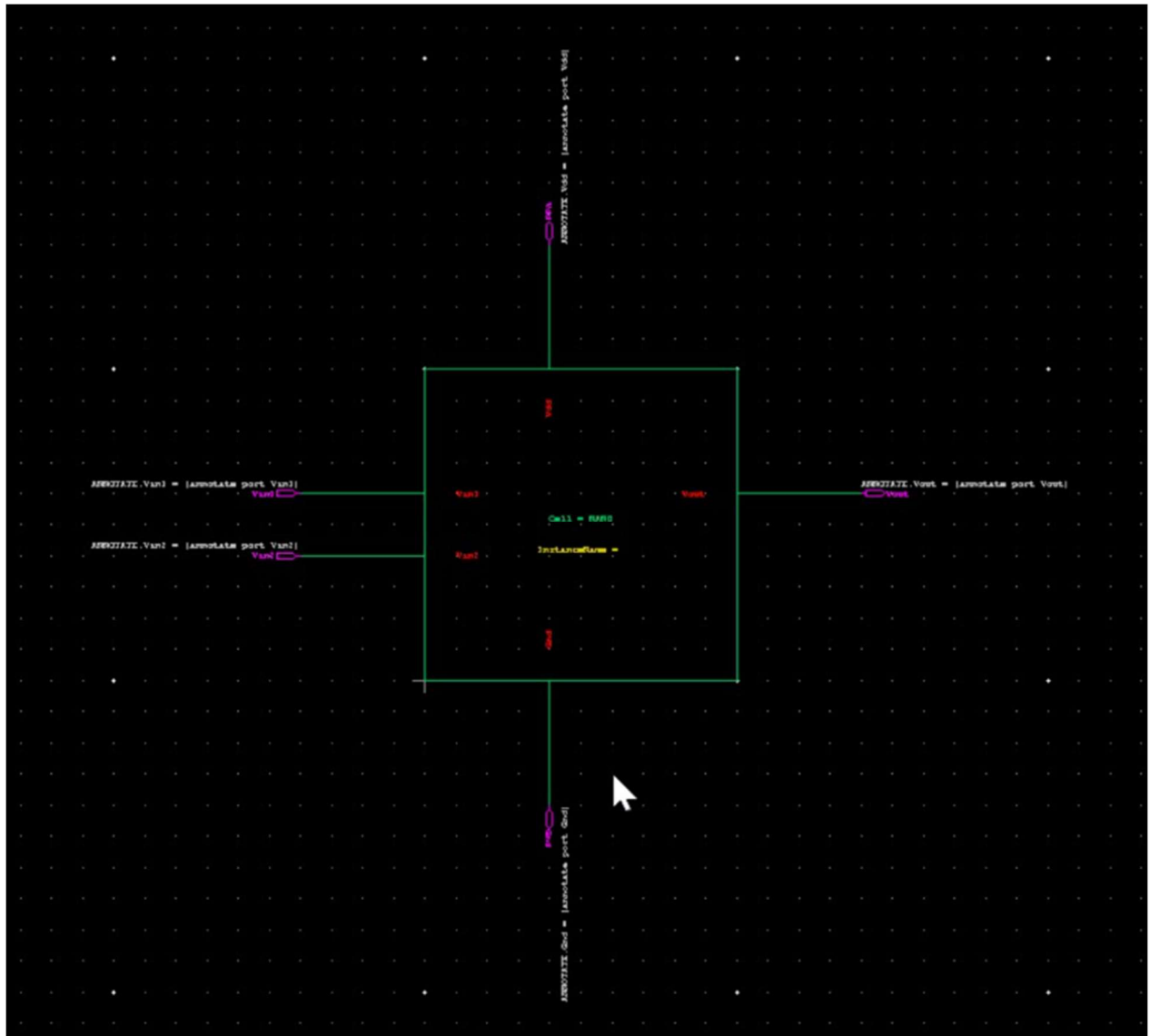


Figure-2 Nand Symbol

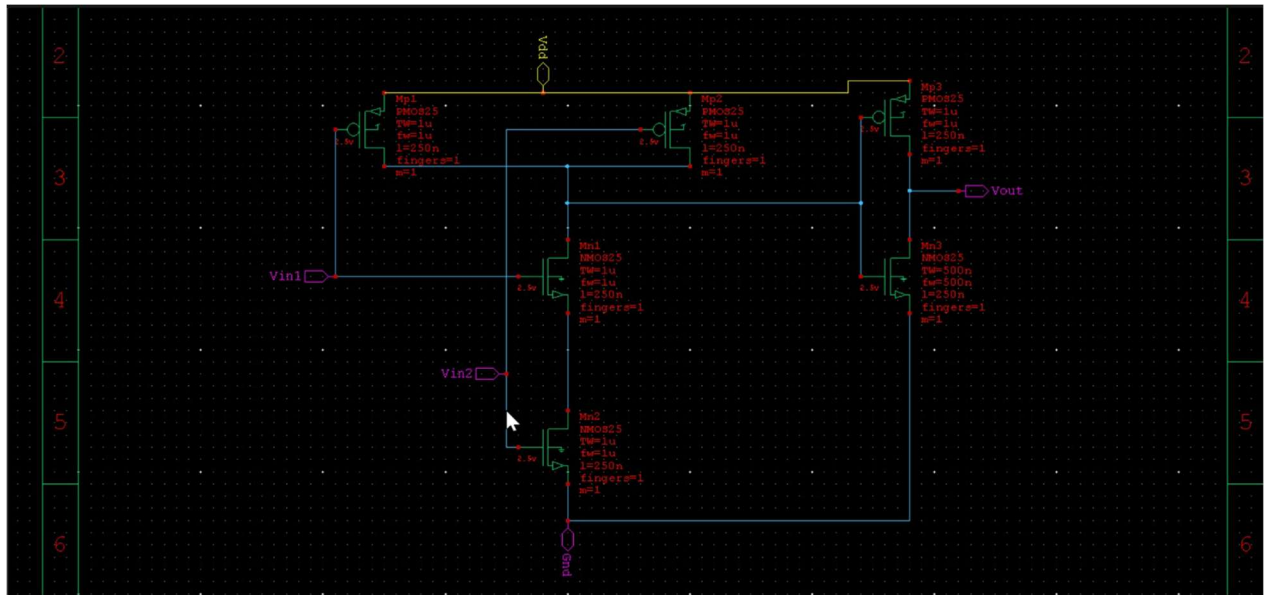


Figure-3 AND Schematic.

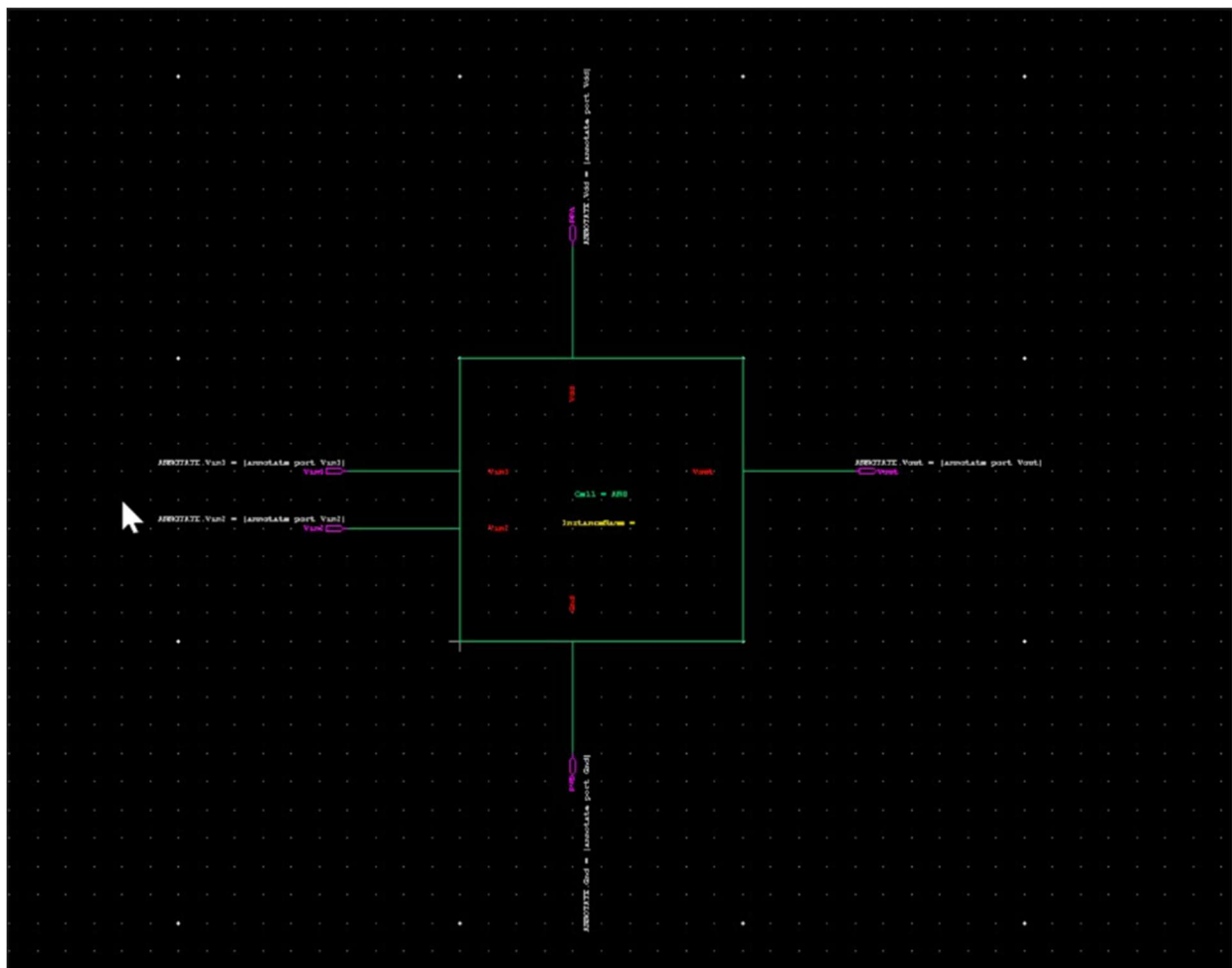
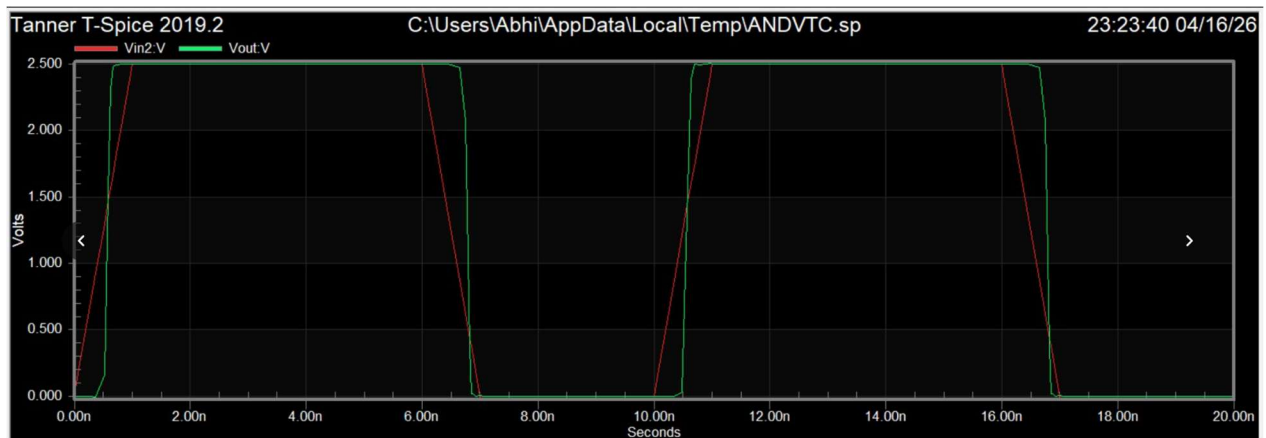
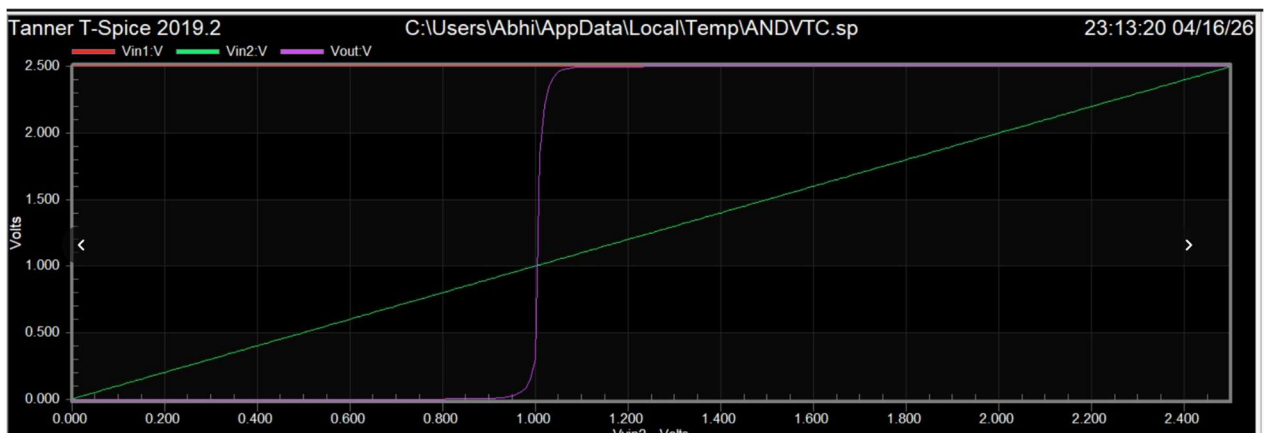


Fig: And Symbol.



VTC of AND Gate For $a=1, b=0-1$

Propagation delays of the gates $T_{pHL}=60\text{ps}$ $t_{pLH}=175\text{ps}$



VTC of AND Gate For $a=0, b=0-1$

Propagation delays of the gates

$t_{pHL}=280\text{ps}$ $t_{pLH}=72\text{ps}$

For OR Gate

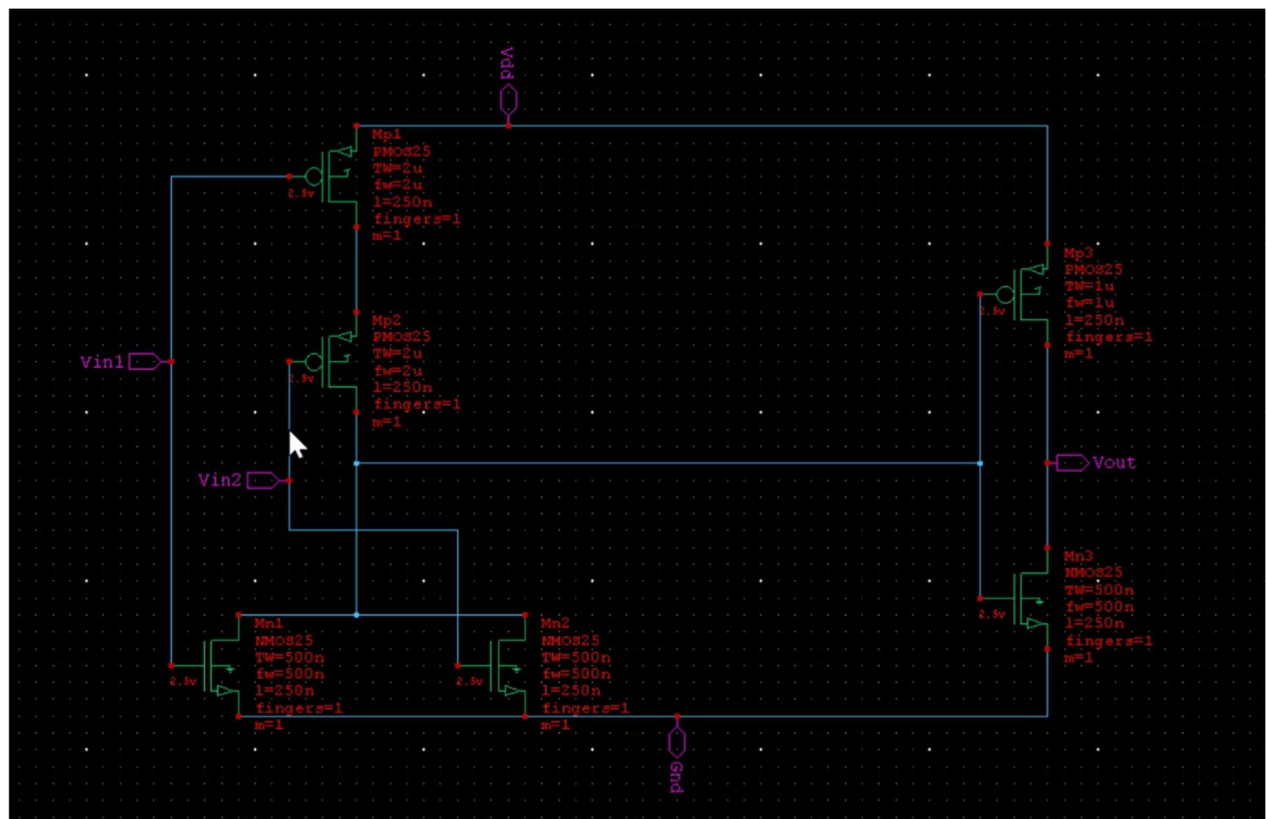


Figure: OR Gate

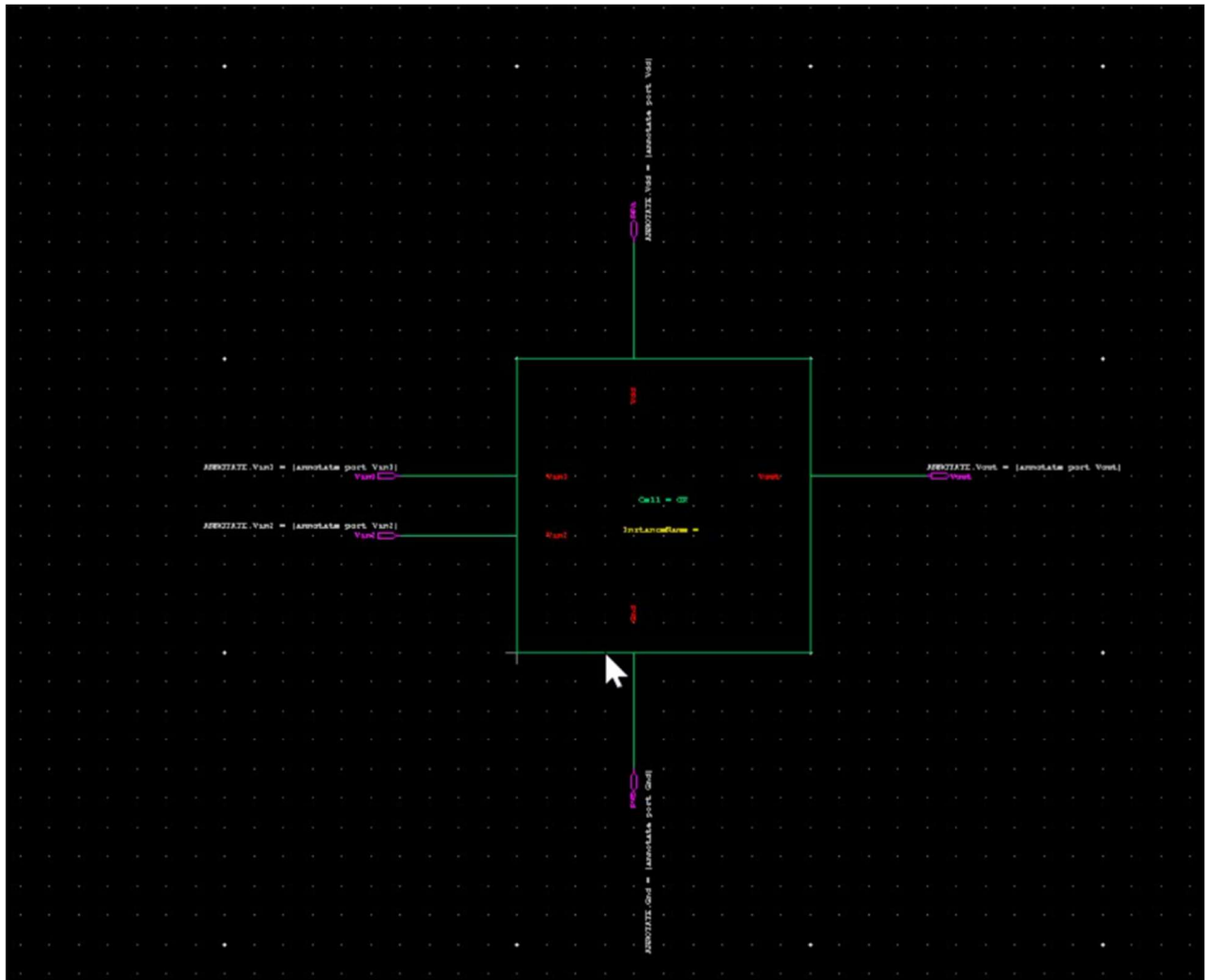
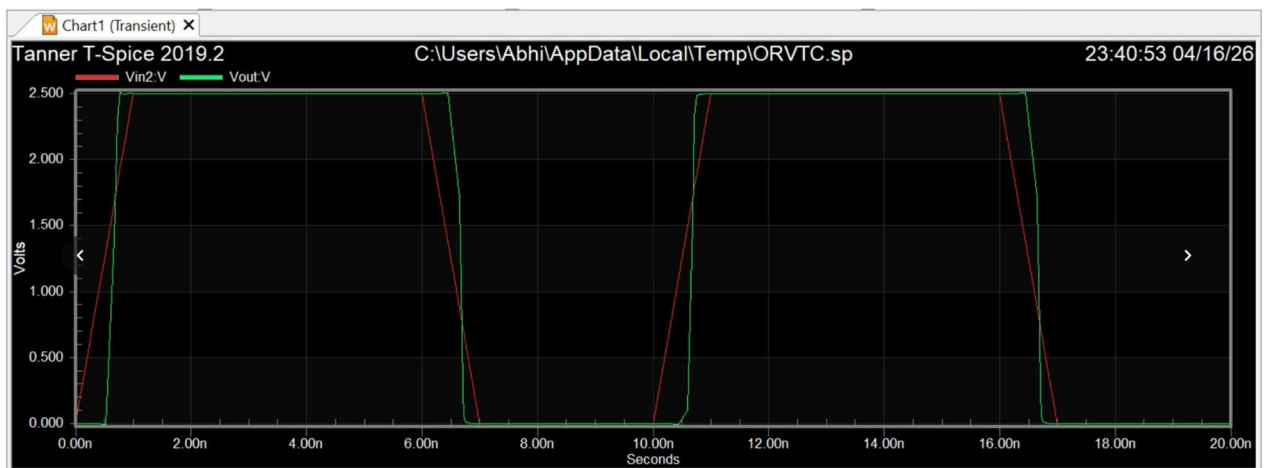


Figure: OR Gate Symbol.



VTC of OR Gate For a=b=0-1

For XOR Gate :

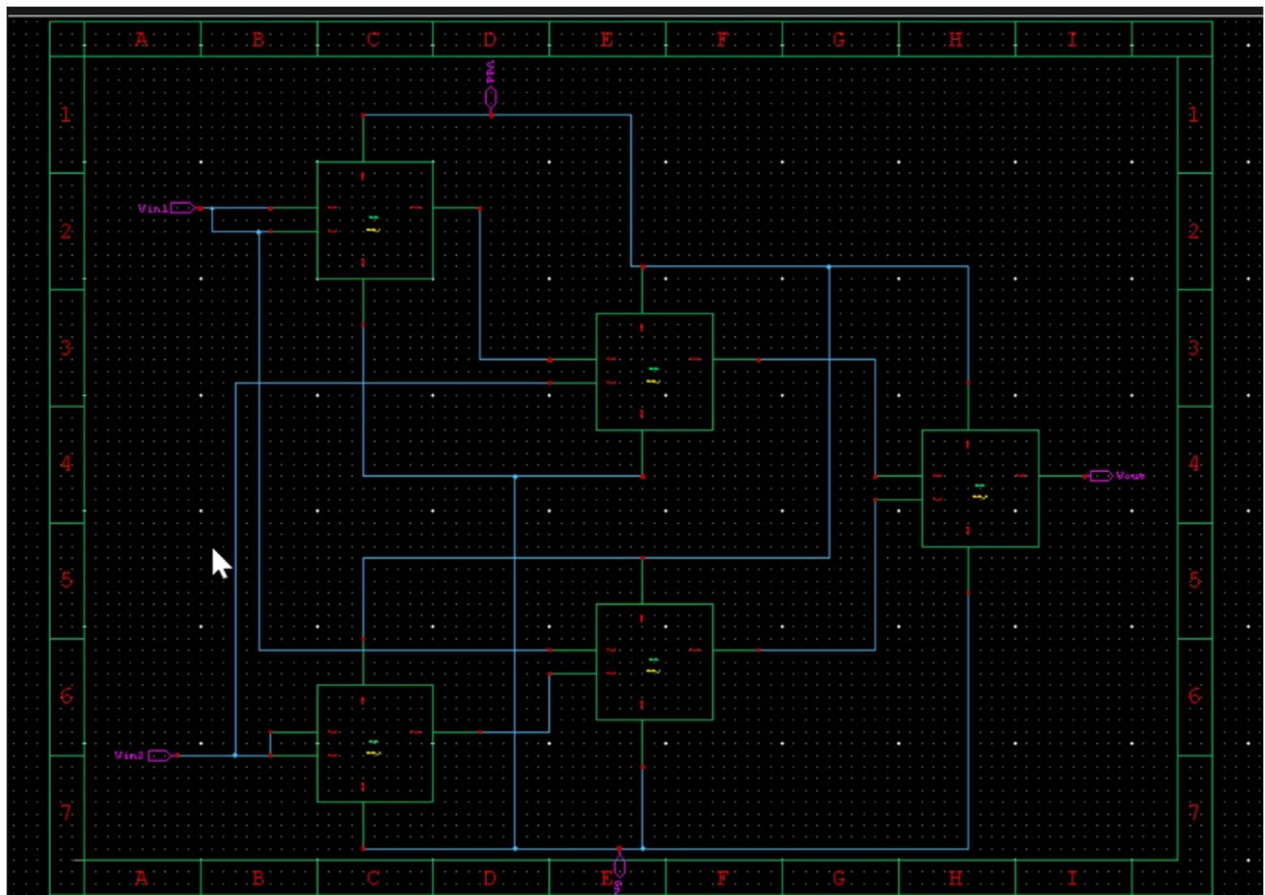
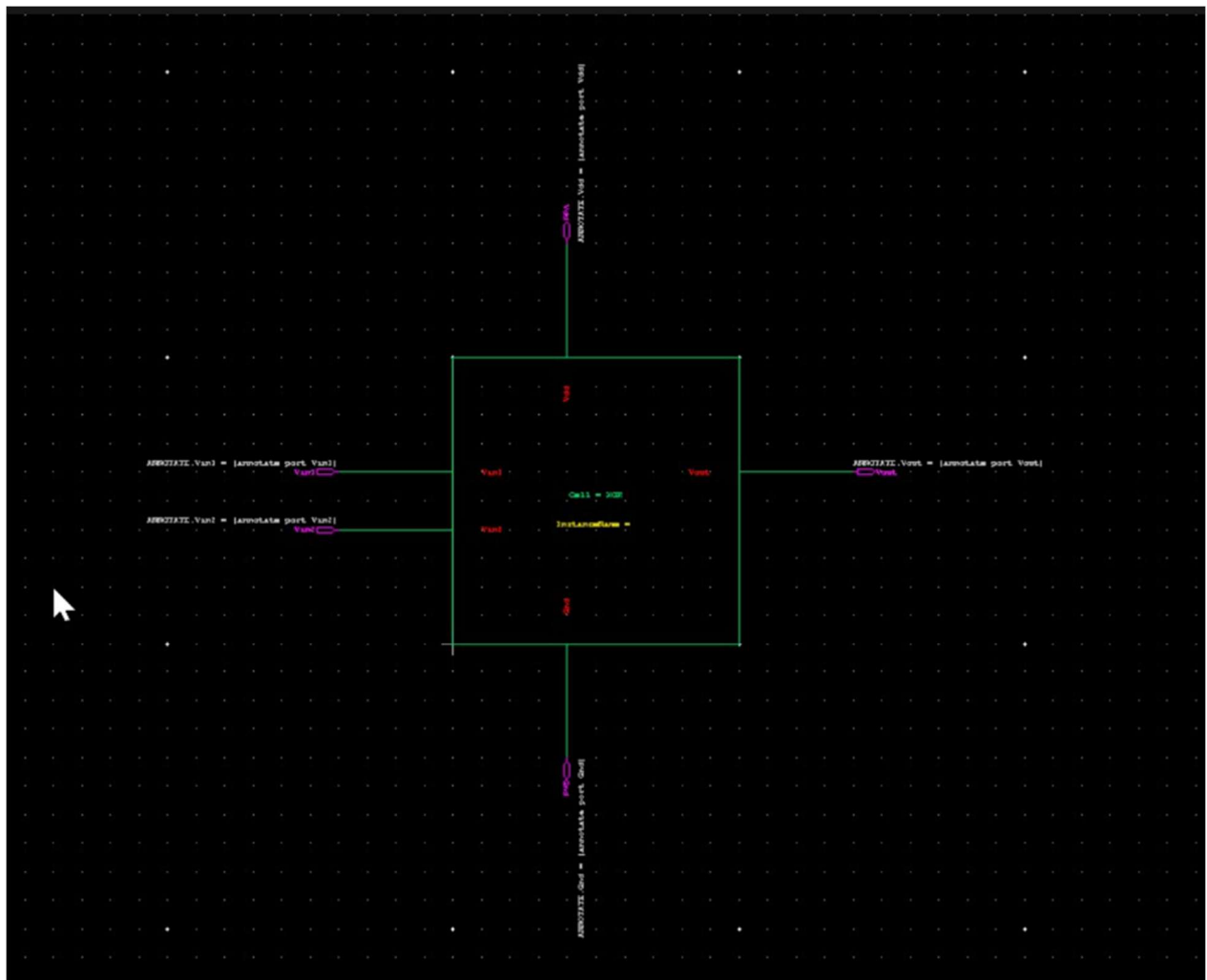
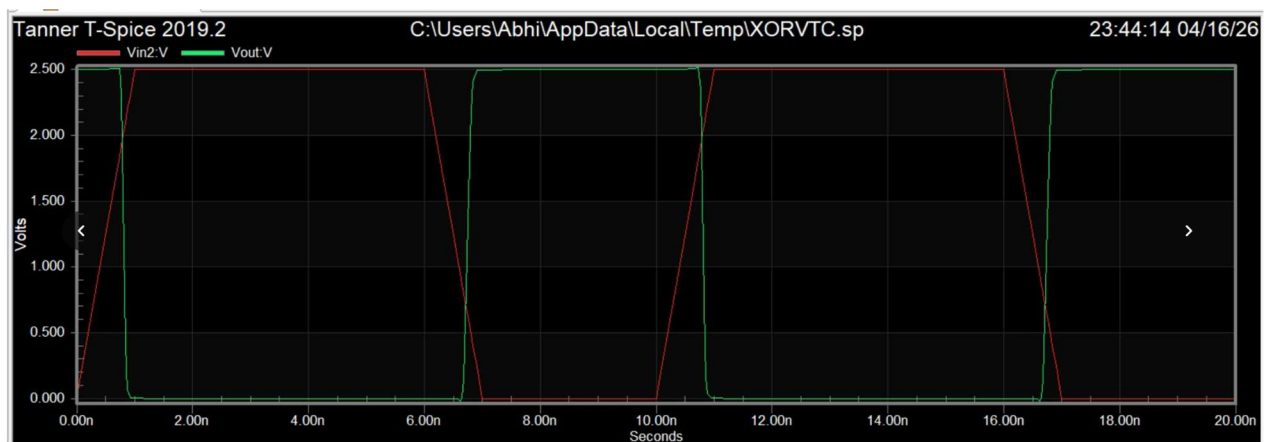


Fig:XOR Gate Schematic.



FigLXOR Gate Symbol.



VTC of XOR Gate a=1,b=0-1

(b) Instantiating the AND, OR, and XOR gates, implement a full adder. Plot the VTC and find out the propagation delays of the full adder assuming that all inputs transition simultaneously.

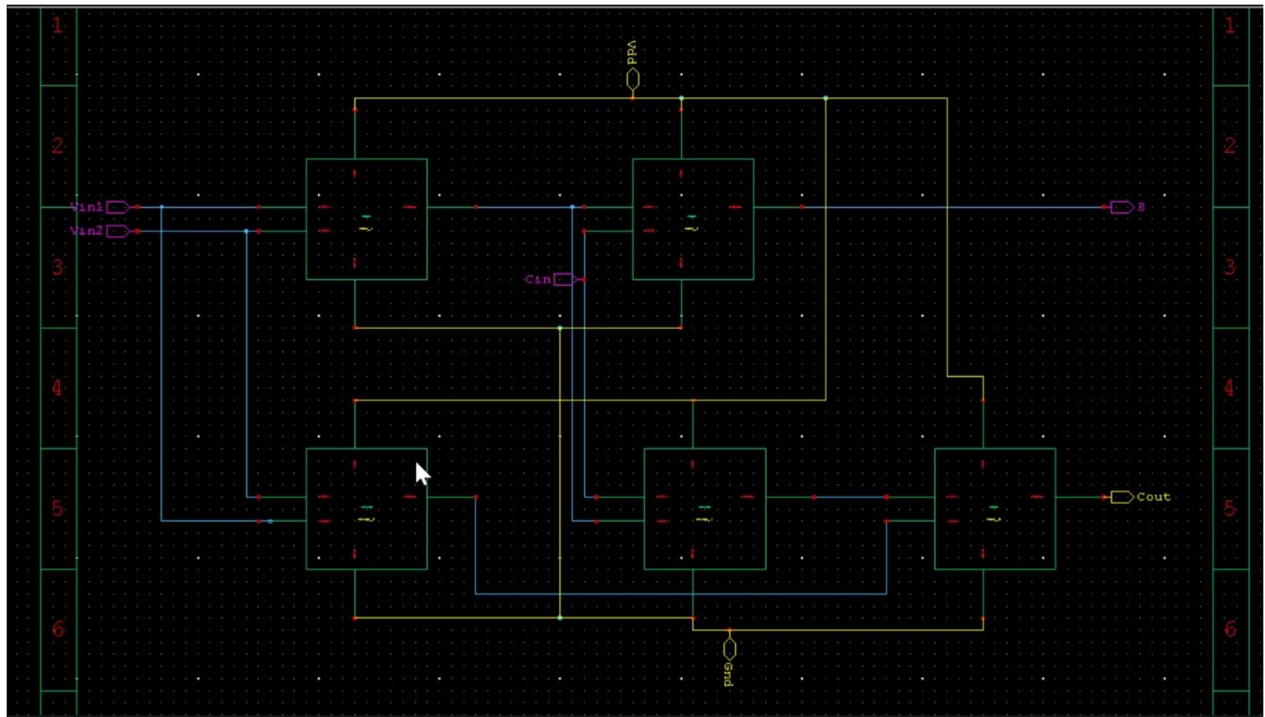


Fig:Full adder schematic .

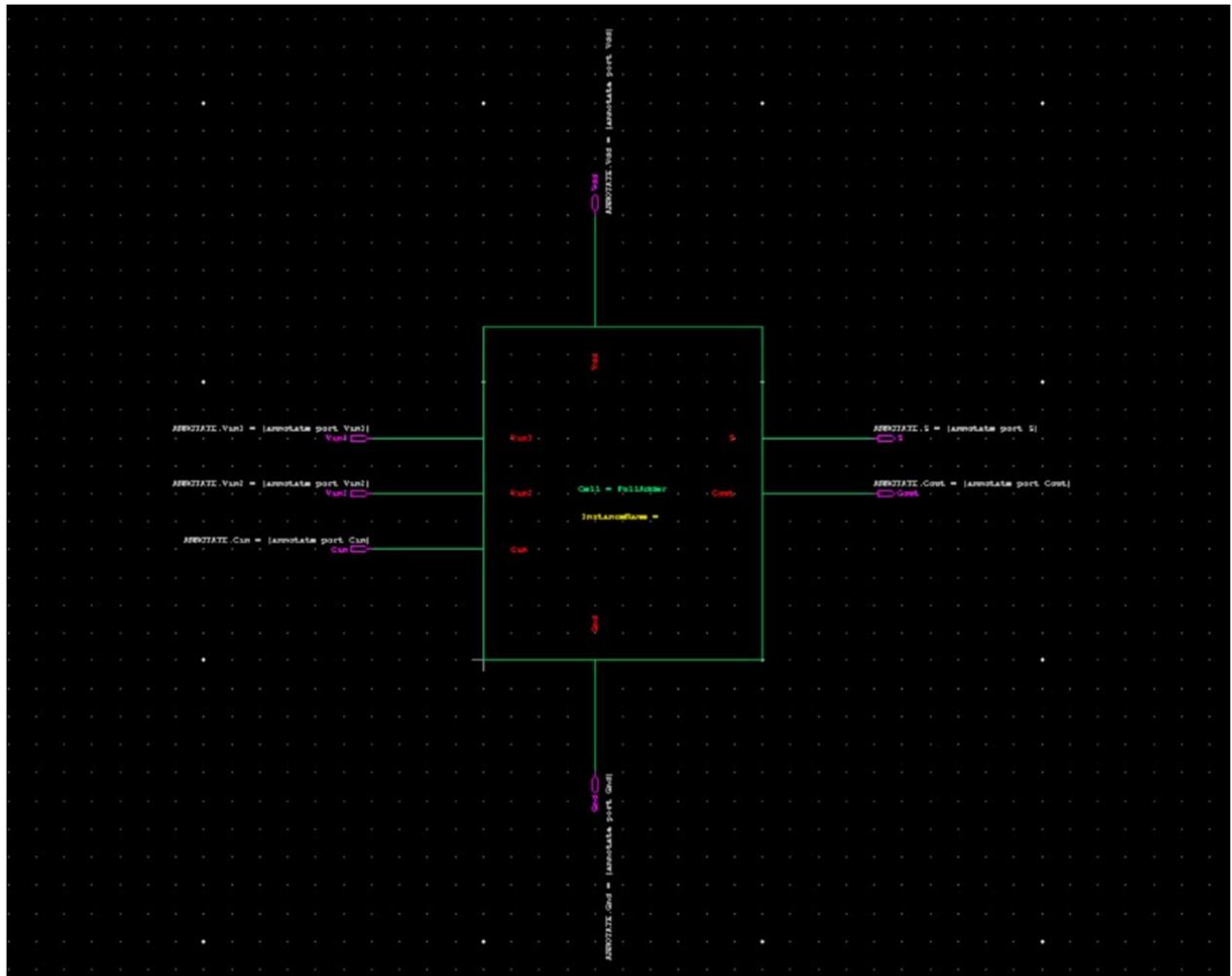


Fig:Full Adder Symbol .

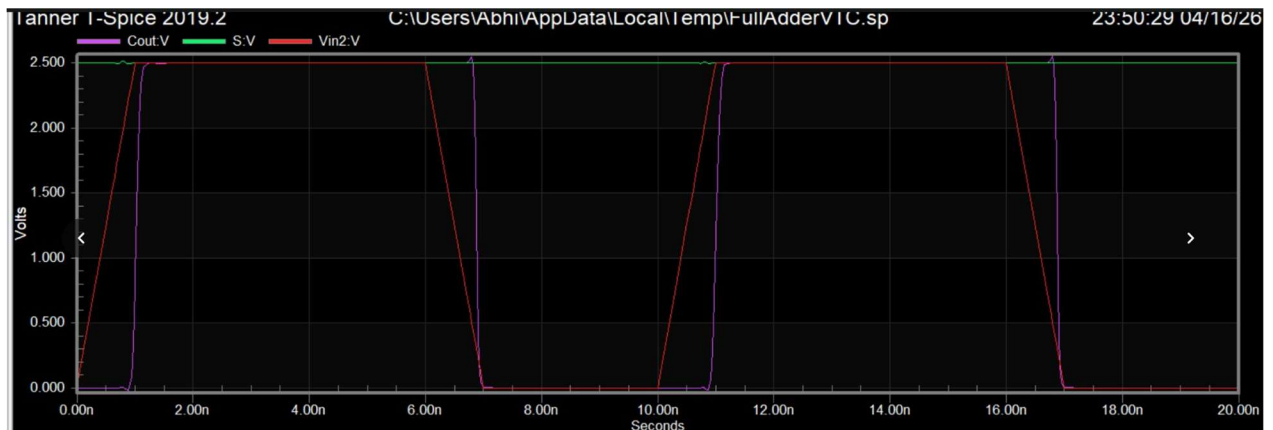


Fig:VTC Of Full Adder A=B=0->1 Cin=1.

Propagation delays of the gates

$t_{pHL}=0.35ns$ $t_{pLH}=0.3ns$

(c) Build a four-bit adder by instantiating the above full adder four times. Demonstrate the functionality through waveforms for 2-3 different input combinations. Estimate the critical path delay through simulations

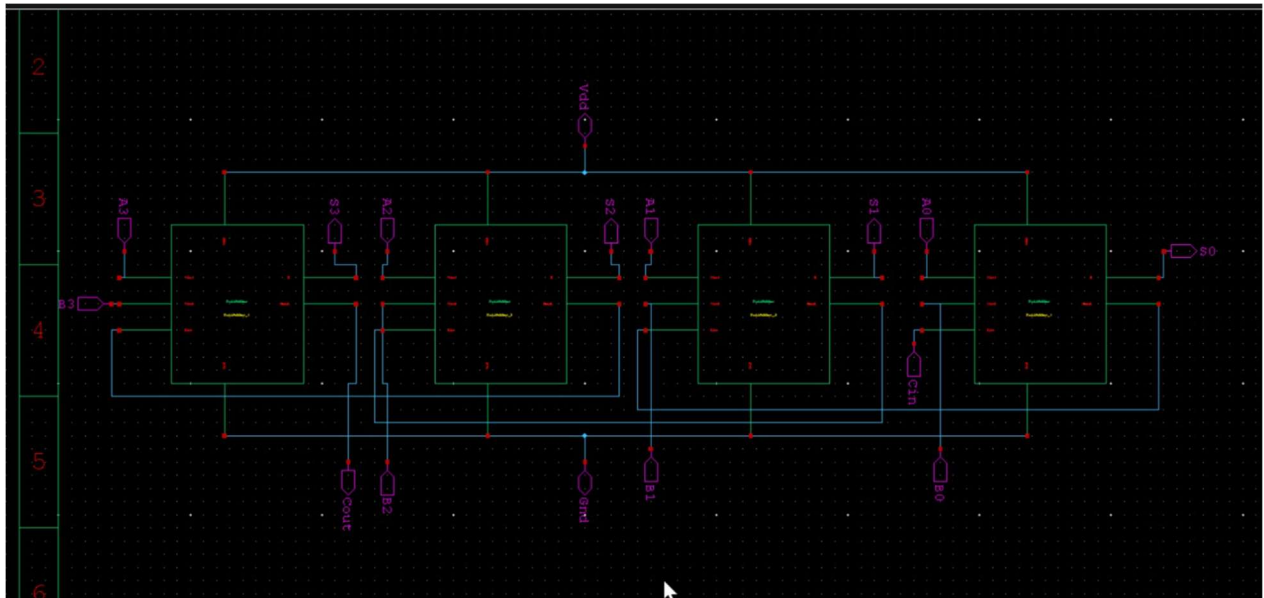


Fig:Schematic of Four bit adder .

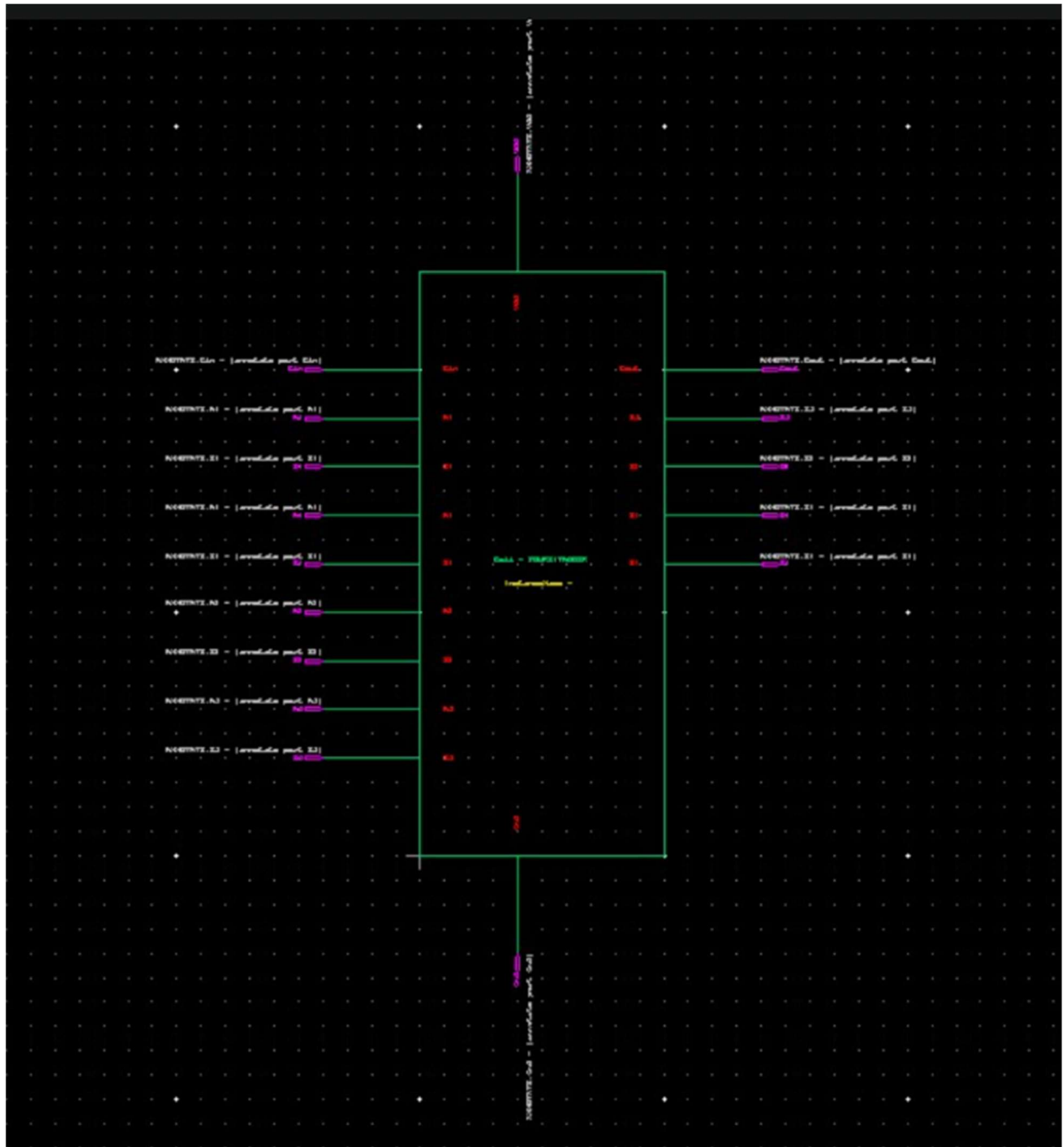
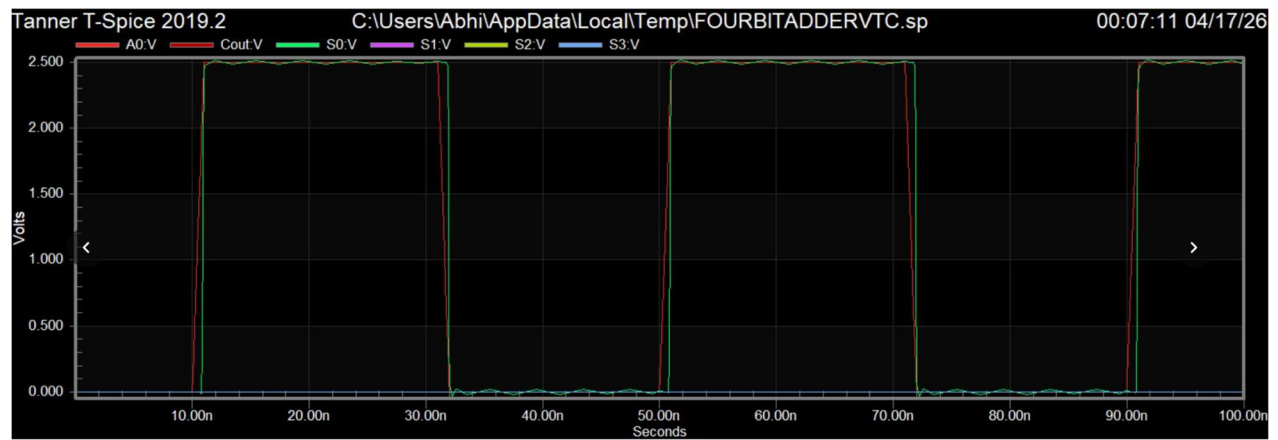


Fig:Symbol of four bit adder .



Case-1 A0 Changing between 0 and 1

Fixed:

$$A1 = 1$$

$$A2 = 0$$

$$A3 = 0$$

$$B = 0111 \rightarrow 0011$$

$$C_{in} = 0$$

Input Condition:

$$A = 0011$$

$$B = 0111 \rightarrow 0011$$

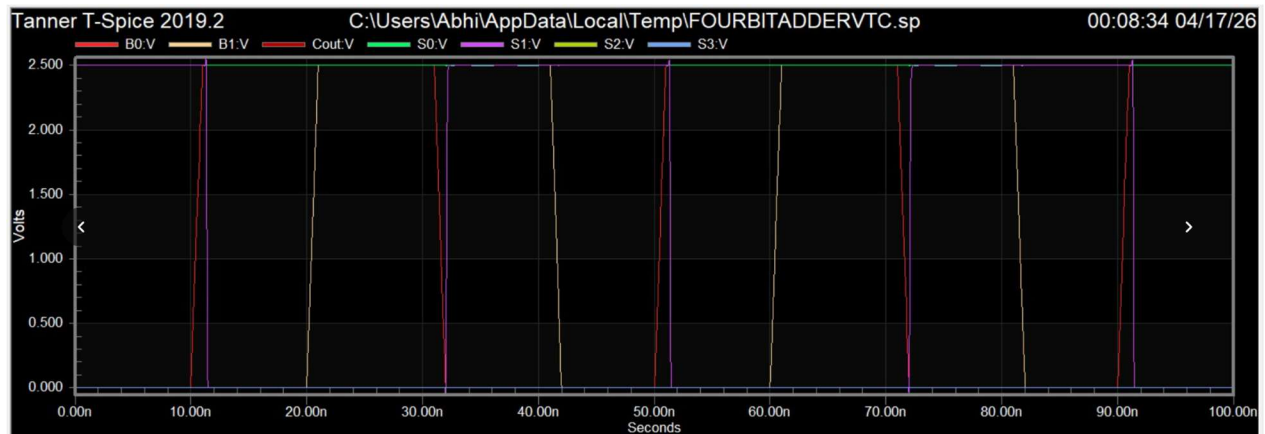
- $C_{in} = 0$

Output

$$S = 1010 \rightarrow 0110$$

$$C_{out} = 0$$

Critical path delay is $t_{cd} = 668.5213\text{ps}$



Case-2

B0 → Pulse

B1 → Pulse

Fixed:

A0 = 1

A1 = 1

A2 = 0

A3 = 0

Cin = 0

Input Condition:

A = 0011

B = 0000 → 0011

Output

S = 0011 → 0110

Cout = 0

Critical path delay is $t_{cd} = 523.7354\text{ps}$

(d) Implement an array multiplier by instantiating the four-bit adder and AND gates as shown in Fig. 1. Demonstrate the functionality through waveforms for 2-3 different input combinations. Estimate the critical path delay through simulations. Calculate the power consumption and the transistor area for this implementation.

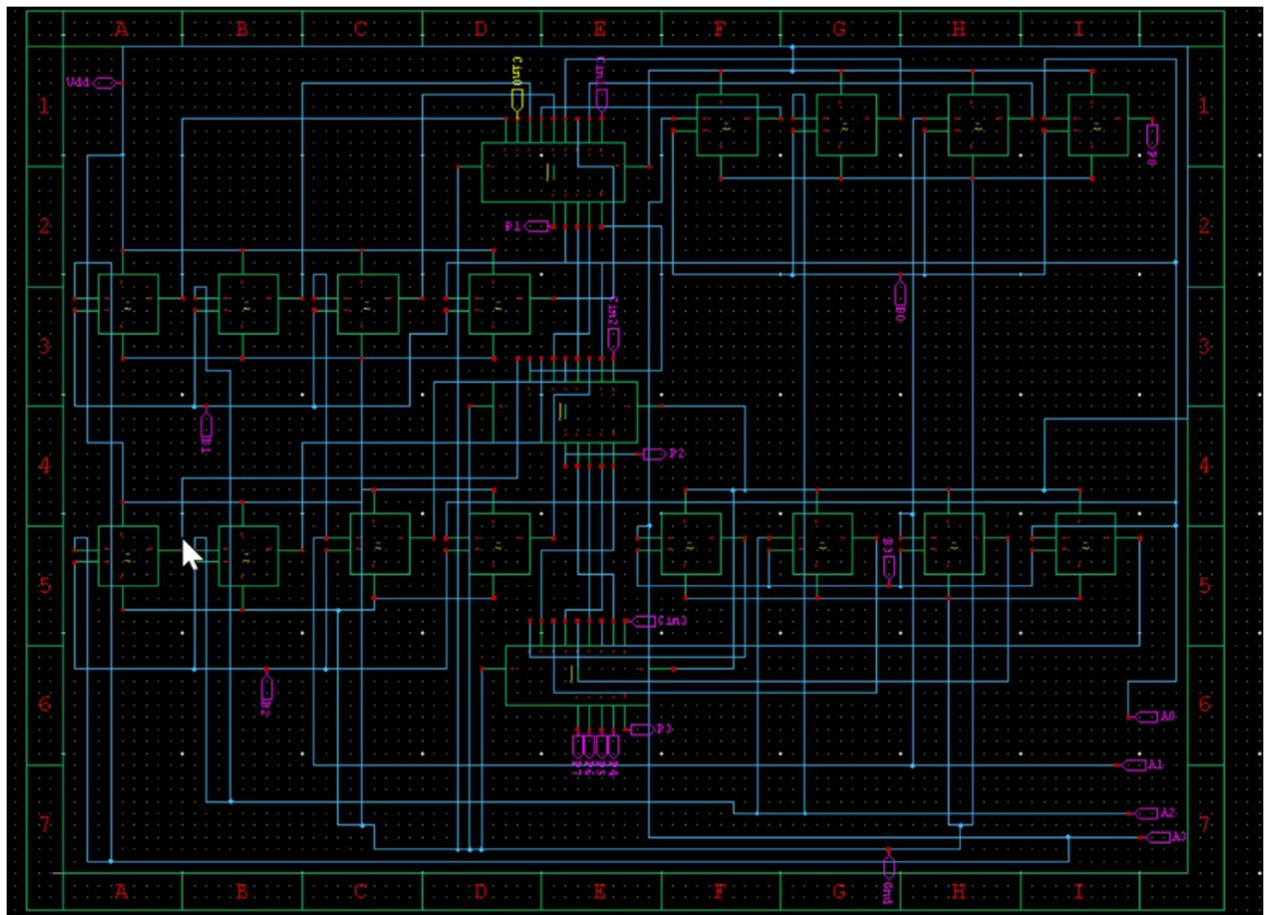


Fig:Schematic of Four bit array multiplier .

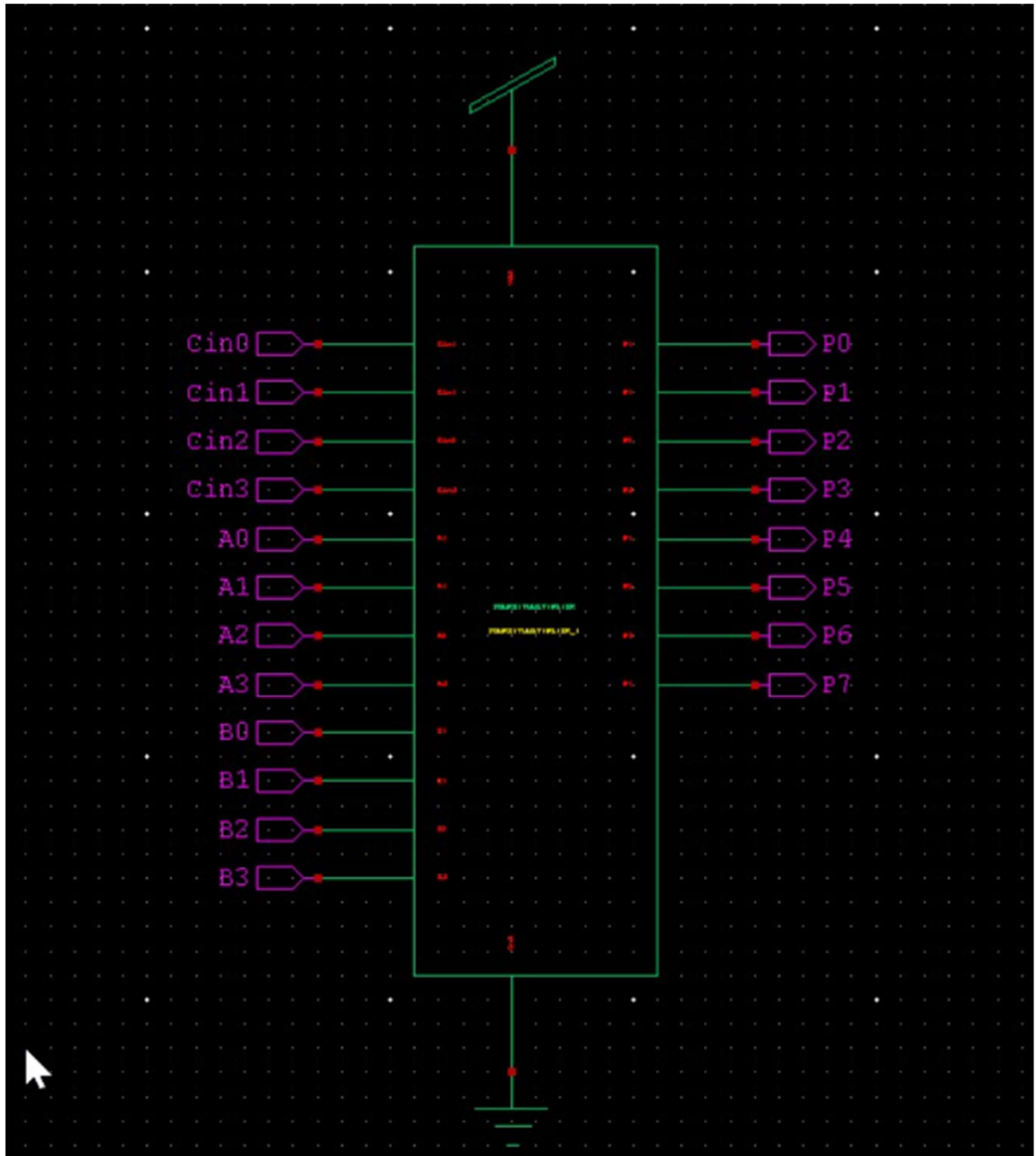


Fig:SymbolVTC of Four bit array multiplier .

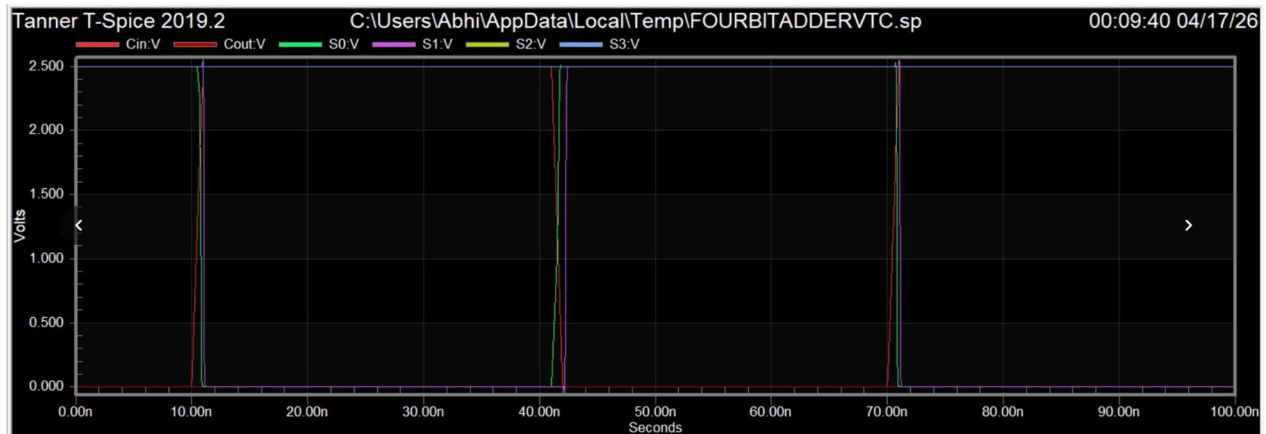


Fig: Four bit multiplier Cin Transistion.



Case 1

Changing:

Ao changing

Fixed:

B = 0011

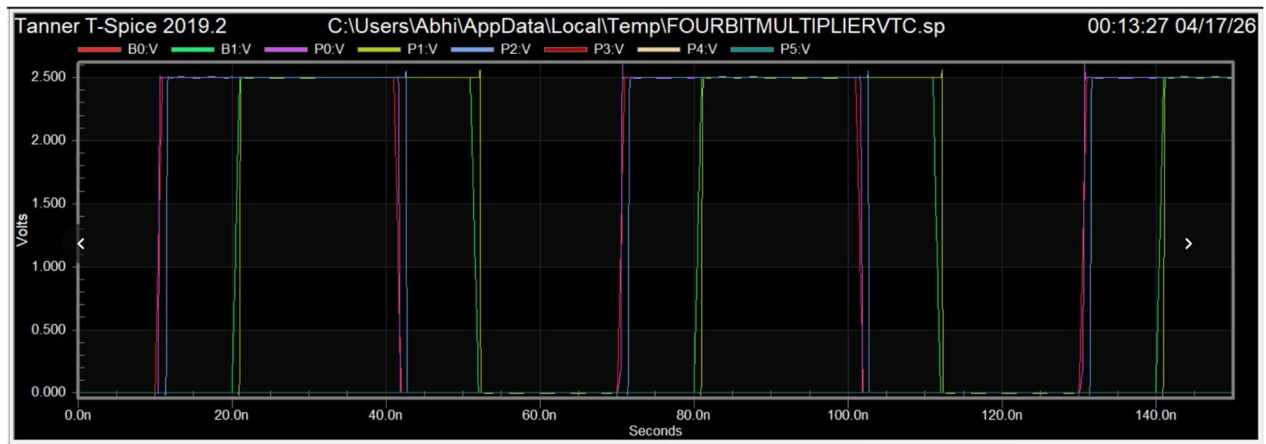
A1 A2 A3 = 0

Carry inputs = 0

Conditions:

A * B only A0 is changing

A0	B0	Output
0	0011	0000
1	0011	0011



Case 2:

Changing:

B0

B1

Fixed:

A = 0101

Multiply:

$0101 \times B$

B changes:

0000

0011

A	B	Output
0101	0000	0000
0101	0011	1111